

[YIJC/2025/Q5]

- 5 A recreational sports club in the college organises training sessions for three new sports and encourages student members to participate regularly.

The three new sports are Lawn Bowls, Pickleball and Tchoukball.

The attendance, the sessions' information and members' data are stored in a relational database.

The database `sportsclub.db` has the following tables:

Member:

- `Id` – unique code for each member, for example 'M2503'
- `Name` – the name of the member
- `Class` – the class of the member

Session:

- `Id` – unique integer code that is auto-generated for each session, for example 2501
- `Date` – the date of the session, for example '2025-03-26'
- `Sport` – the sport training conducted during the session

Attendance:

- `MemberID` – the unique member code
- `SessionID` – the unique session code
- `Status` – the attendance of the member for the session, i.e. '1' for present and '0' for absent.

Task 5.1

A training session (session id: 2504) was conducted on 14th Feb 2025 for the sport Lawn Bowls and the attendance was stored in a text file `SessionID_2504_ATTENDANCE.TXT`.

Write Python program code to insert the session information and the attendance records into their respective tables in the database.

Save your Python program as `Task5_1_<your name>.py`.

[4]

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A web application is to be designed to allow the teacher to view and update the attendance.

The default webpage `index.html` will display the following menu:

Sports Club Attendance

1. [Members List](#)
2. Choose a sport to display the attendance:
3. [Absentees List](#)
4. [Mark Attendance](#)

Task 5.2

Write a Python program and the necessary files to create a web application to display **only the first option** (1. `Members List`) in the menu for the teacher to view the list of members in the Sports Club.

The program should return an HTML document `members.html` that enables the web browser to display a table containing the Class and Name of the members.

Save your Python program as:

`Task5_2_<your name>.py`

with any additional files / subfolders in a folder named:

`Task5_2_<your name>`

Run the web application.

Save the webpage output as:

`Task5_2_<your name>_output.html`

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Task 5.3

Modify your Python program and the default webpage `index.html` to include the **second option** (2. Choose a sport ...) in the menu for the teacher to select to view the attendance for a particular sport.

The program should return an HTML document `attendance.html` that enables the web browser to display a table containing the following data:

- session dates
- member names
- attendance statuses.

Save your Python program as:

`Task5_3_<your name>.py`

with any additional files / subfolders in a folder named:

`Task5_3_<your name>`

Run the web application.

Save the webpage output as:

`Task5_3_<your name>_output.html`

[5]

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Task 5.4

Modify your Python program and the default webpage `index.html` to include the **third option** (3. Absentees List) in the menu for the teacher to select to view the list of absentees for all the sessions.

The program should return an HTML document `absent.html` that enables the web browser to display a table containing the following data:

- member names
- member classes
- session dates

The data in the table should be sorted by the members' names.

Save your Python program as:

`Task5_4_<your name>.py`

with any additional files / subfolders in a folder named:

`Task5_4_<your name>`

Run the web application.

Save the webpage output as:

`Task5_4_<your name>_output.html`

[3]

During the training session, the teacher will mark the attendance of the members.

An example of the webpage is shown below:

Members List for Attendance Marking

Session Date (yyyy-mm-dd):

Sport:

Please Select

Class	Name	Present (?)
25A21	Yi Xuan	☐
25A21	Ravi	☐
25A21	Zhi Yong	☐
25S31	Aisyah	☐
25S31	Nurul	☐
25S31	Jun Hao	☐
25S32	Azman	☐
25S32	Jia Hui	☐
25S32	Anjali	☐

Submit

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Task 5.5

Modify your Python program and the default webpage `index.html` to include the **fourth option** (4. Mark Attendance) in the menu for the teacher to view the members' list and mark the attendance.

The program should return an HTML document `mark.html` that enables the web browser to display the following:

- a text box to enter the session's date in the `yyyy-mm-dd` format
- a list to select one of the three sports
- a table with three columns:
 - member classes
 - member names
 - a column for the teacher to indicate if the member is present for the training session.

The data in the table should be sorted by the members' classes.

The submitted session information and the attendance records should be updated into their respective tables in the database.

After updating the database, the program should return an HTML document `success.html` that enables the web browser to display the following message:

"Attendance Updated."

Save your Python program as:

`Task5_5_<your name>.py`

with any additional files / subfolders in a folder named:

`Task5_5_<your name>`

Run the web application.

Save the attendance taking webpage output as:

`Task5_5_<your name>_output.html`